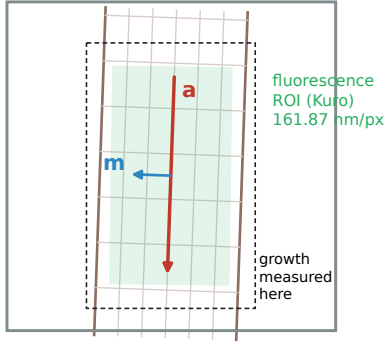


b The growth model: seven numbers per snapshot

a Two cameras, one sample plane

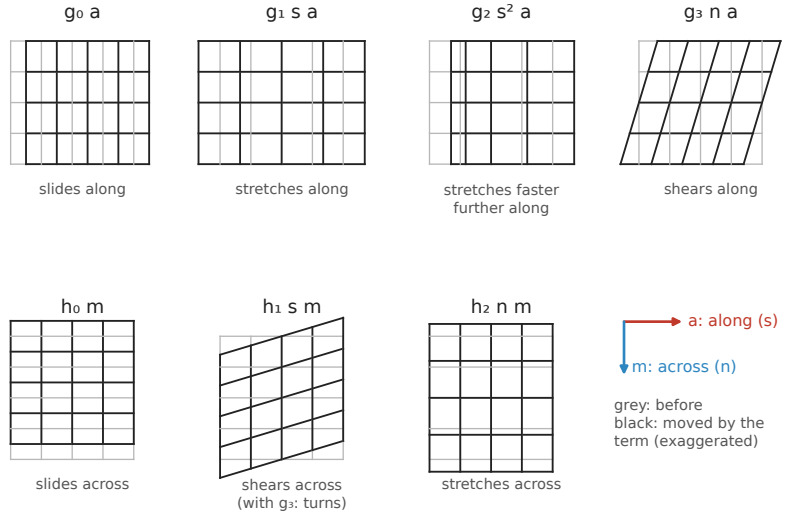
white light (Basler): 166 μm , 81.28 nm/px



camera map C: $q = J b + T$, $J = 0.502 R(0.80^\circ)$

b white-light px, q fluorescence px (Argo-SIM, 5-13 nm)

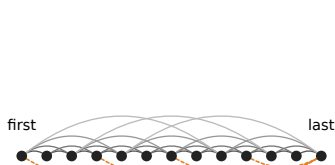
a: the root axis, from the direction of the walls



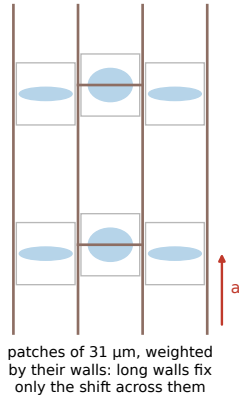
$$\Phi_k(p) = p + (g_0 + g_1 s + g_2 s^2 + g_3 n) a + (h_0 + h_1 s + h_2 n) m$$

Snapshot k onto the last one's geometry: an affine map (6 numbers) plus the one quadratic term a growing root needs. s, n: position along and across the root axis, from the field centre.

c Estimation, on the white-light snapshots

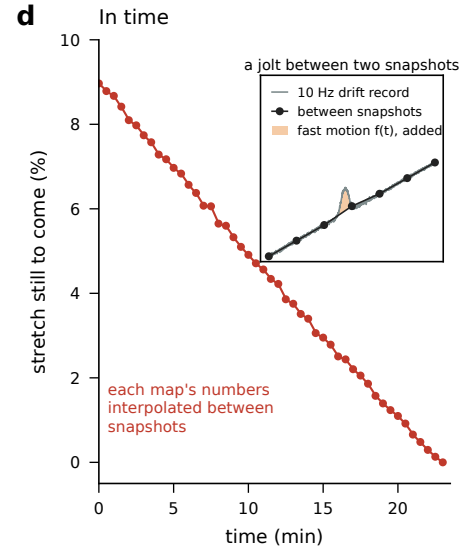


snapshots every 30 s; pairs 1, 2, 4, 8, 16 apart, and each against the last (orange): every map tied to the last by many paths

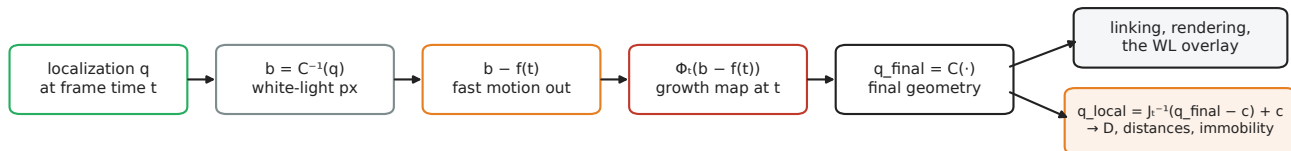


patches of 31 μm , weighted by their walls: long walls fix only the shift across them

- 1 rough maps: neighbours chained
- 2 every snapshot warped into the last one's geometry
- 3 every pair registered, patch by patch: $r_{jk} = e_j - e_k$ (e: what a map has wrong)
- 4 one weighted, robust fit over all pairs: every e_k
- 5 maps updated; repeated until < 0.05 px (4 nm)



e One localization, corrected



Every localization lands where its piece of tissue is at the end. For D, the local stretch still to come (j_t) is taken back out: linked in the final geometry, a step between neighbouring frames would be stretched by it, and D inflated by $(1 + \epsilon)^2$.